

SMART HOSPITAL ASSISTANT ROBOT FOR AUTONOMOUS MEDICINE DELIVERY

CSE461: Introduction to Robotics
BRAC University
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ABSTRACT

The Smart Hospital Assistant Robot is designed to automate routine medicine delivery within a hospital environment. The system uses an ESP32-based mobile platform integrated with line-following sensors, ultrasonic obstacle detection, ESP32-CAM, Wi-Fi communication, and servo-controlled medicine compartments. Delivery tasks are assigned through a PC-based control system. The robot autonomously follows a predefined path from the medicine station to a selected patient location while detecting obstacles along the way. At the destination, the ESP32-CAM provides visual information for patient or destination verification. Medicine is released only after successful verification, helping reduce incorrect or unauthorized delivery. After completing the task, the robot returns to the medicine station and enters standby mode. The project demonstrates a practical, low-cost application of robotics, embedded systems, sensing, communication, and automation in smart healthcare.

INTRODUCTION

- **Motivation:** Reduce repetitive medicine-delivery workload of healthcare workers.
- **Problem:** Manual delivery may cause delays, human error, and unnecessary physical contact.

OBJECTIVES

- Autonomous medicine delivery
 - Obstacle detection
 - Destination verification
 - Secure medicine transport
 - Automatic return to base
- Application: Hospitals, clinics, and nursing homes.

FUNCTIONALITIES

Mechanical Functionality

- Uses a 2WD chassis with DC gear motors for movement.
- Follows a predefined floor path.
- Carries medicine inside a dedicated medicine compartment.
- Servo motors lock the compartment during transportation.
- The compartment unlocks only after successful verification.
- The robot returns to the medicine station after completing delivery.

Software Functionality

Standby → Receive Delivery Task → Lock Medicine → Navigate → Detect Obstacles → Reach Destination → Capture Image → Verify Destination → Unlock Medicine → Complete Delivery → Return to Base → Standby

Navigation Functionality

Line Following

The 5-array sensor continuously detects the predefined floor line and sends navigation information to the ESP32.

Obstacle Detection

The ultrasonic sensor continuously measures the area in front of the robot. Obstacle detected → Robot stops

Path becomes clear → Robot resumes movement

This allows the robot to travel autonomously while reducing the risk of collision.

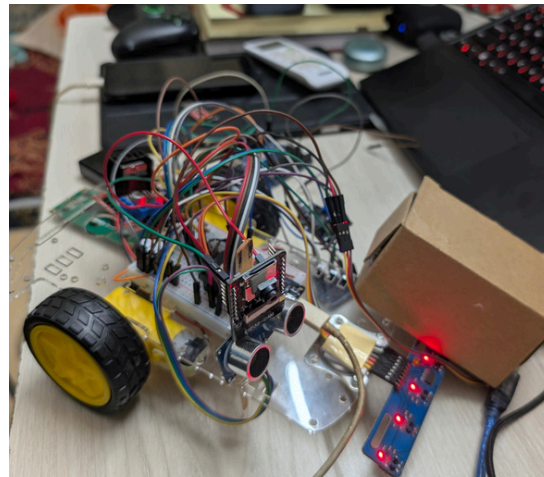
Software-Hardware Communication

Wi-Fi: Communication between the ESP32 and PC-based system for: Delivery task assignment, Robot monitoring, Verification-related processing

GPIO / PWM: Used to control: DC motors, Motor driver, Servo motors, Buzzer

I2C: Used for communication with the I2C display.

SYSTEM OVERVIEW



ESP32 | ESP32-CAM | Ultrasonic Sensor | 5-Array Sensor | L298N |
DC Motors | Servo Motors | Medicine Box | Battery | 2WD Chassis

OVERALL PROCESS

MEDICINE STATION → NAVIGATION → PATIENT BED
→ VERIFICATION → DELIVERY → RETURN

IMPLEMENTATION

CONTROL SYSTEM

BATTERY → VOLTAGE

REGULATOR → ESP32

ESP32 connects to:

- 5-Array Sensor — Line following
- Ultrasonic Sensor — Obstacle detection
- L298N + DC Motors — Movement
- Servo Motors — Compartment control
- ESP32-CAM — Verification
- LCD + Buzzer — Status/notification
- Wi-Fi + PC — Task assignment

Hardware

- ESP32
 - ESP32-CAM
 - 5-Array Sensor
 - Ultrasonic Sensor
 - L298N Motor Driver
 - 2WD Chassis
 - SG90 Servo *2
 - DC Motors
 - I2C Display
 - 5V Buzzer
 - Battery
 - LN2596
- Estimated Cost: Tk. 4840

Software

- Arduino Uno

TASK EXECUTION FLOW

1. Standby
- ↓
2. Task Assigned from PC
- ↓
3. Medicine Loaded
- ↓
4. Compartment Locked
- ↓
5. Follow Line
- ↓
6. Obstacle Detection
- ↓
7. Reach Destination
- ↓
8. Capture Image
- ↓
9. Verify Patient / Destination
- ↓
10. Unlock Compartment
- ↓
11. Medicine Collected
- ↓
12. Return to Medicine Station
- ↓
13. Standby

APPLICATIONS & COMPITORS

APPLICATIONS

- HOSPITAL MEDICINE DELIVERY
- CLINICS AND NURSING HOMES
- INDOOR HEALTHCARE LOGISTICS

ADVANTAGES

- LOW-COST
- AUTONOMOUS
- SECURE COMPARTMENT
- DESTINATION VERIFICATION
- AUTOMATIC RETURN

LIMITATIONS

- REQUIRES MARKED PATHS
- DEPENDS ON WI-FI
- CAMERA AFFECTED BY LIGHTING
- DESIGNED FOR CONTROLLED ENVIRONMENTS

CONCLUSION

The proposed robot combines autonomous navigation, obstacle detection, wireless communication, destination verification, and secure medicine delivery in a low-cost healthcare robotic system.

REFEFENCES

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FUTURE WORK

- LiDAR/SLAM navigation, better computer vision, automatic charging, and multiple medicine compartments.